The Biodiversity Impact of Agricultural Trade Liberalization*

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[PRELIMINARY AND INCOMPLETE]

Abstract

Agriculture is identified as a major contributor to global biodiversity decline. This paper examines the role of agricultural trade in governing bird biodiversity losses, through changes at both the extensive and intensive margins of land-use. To do so, I combine a multi-country general equilibrium model of trade, which accounts for the use and trade of agrochemicals, with a species-area and a population dynamics model. I run a set of counterfactual scenarios to estimate the impact of historical trade policies on: (i) long-term bird species richness in 700 ecoregions of the world, and (ii) local abundance of common birds in the United States. Preliminary results on species richness suggest that trade liberalization limits losses globally, but at the expense of southern biodiversity hotspots. As birds provide essential ecosystem services, future work will focus on using these estimates to predict changes in agricultural productivity and, subsequently, in food security and welfare.

Keywords: agriculture, trade policy, biodiversity, land-use, birds
JEL Codes: F18, Q15, Q17, Q57

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1 Introduction

The current biodiversity crisis is unprecedented in human history. More than 48,600 species are classified as threatened with extinction, and 16% of species assessed as being at lower risk nevertheless show a downward trend in their populations (lUCN, 2025). Owing to its critical role in land-use change, agriculture is identified as the primary threat to biodiversity. On the extensive margin, the expansion of pasture and cropland over pristine areas leads to habitat loss and fragmentation. On the intensive margin, the conversion from subsistence farming to intensive agriculture further erodes ecosystems, increasingly exposing wildlife to (sub)-lethal doses of agrochemicals.

Public discourse frequently points to agricultural trade liberalization as a pincipal driver of these land-use transformations. In theory, liberalizing both input and output markets influences the extensive and intensive margins in opposite directions. At the global scale, reduced trade barriers can shrink the total area required for food production by enabling a more efficient allocation of crops through specialization, and by fostering productivity gains via technology adoption. Conversely, access to cheaper inputs, such as machinery, fertilizers and pesticides tends to expand global land-use at the intensive margin. At the local level, trade liberalization can alter farm profitability, prompting either further conversion of natural habitats or the abandonment of agricultural lands. Overall, the net effects on biodiversity are ambiguous and likely to be spatially heterogenous. Yet, they remain undetermined.

This paper fills this gap by investigating the role of agricultural trade in governing biodiversity losses, considering changes at both the extensive and intensive margins of land-use. Building on the general-equilibrium framework of Farrokhi and Pellegrina (2023) and Gouel and Laborde (2021), I integrate two complementary ecological sub-models to trace the consequences of trade-induced land-use adjustments, focusing on bird species richness and abundance. Counterfactual simulations allow me to map the spatial distribution of biodiversity losses under both historical and hypothetical trade policies. Looking ahead, I intend to extend the analysis by modelling and estimating the feedback loops whereby the biodiversity outcomes generated in these counterfactuals affect agricultural productivity and, subsequently, food security and welfare.

Birds make an excellent focal group for this analysis for several reasons. First, avian populations are undergoing steep declines: the IUCN Red List indicates that 1 in 8 bird species is threatened with extinction, and 73% of those threats are tied to agricultural pressures (lUCN, 2025). Moreover, mounting evidence shows that expansion at the intensive margin of land-use drives bird biodiversity loss (Li, Miao, and Khanna, 2020; Rigal et al., 2023). This impact can occur directly, through ingestion of lethal concentrations of agrochemicals (Lopez-Antia et al., 2015), or indirectly, via reduced food availability, delayed migration, and ultimately lower reproductive success (Eng, Stutchbury, and Morrissey, 2019). Second, birds are highly observable, resulting in exceptionally rich, high-resolution datasets that support spatially explicit modelling. Third, avian communities are widely regarded as robust indicators of overall ecosystem health, meaning that patterns revealed for birds are likely to be informative for other taxonomic groups as well (Padoa-Schioppa et al., 2006). This indicator value stems from birds occupying multiple trophic levels and their sensitivity to subtle environmental changes, including pollution. Consequently, although this paper concentrates on birds, the findings are expected to be generalizable to other ecological taxa. Finally, birds provide essential ecosystem services to agriculture, as summarized in Mariyappan et al. (2023). These services include supporting functions such as seed dispersal (Hougner, Colding, and Söderqvist, 2006) and pollination (Rangaiah, Raju, and Rao, 2004), as well as regulating functions like insect pest control (Mols and Visser, 2007; Karp et al., 2013) and rodent control (Peleg et al., 2018). Consequently, trade-induced shifts in bird biodiversity are expected to influence agricultural productivity and, in turn, generate feedback effects on land-use patterns, trade flows, and welfare in the future—an interconnection that this paper seeks to examine.

I draw on the general equilibrum model of trade from Farrokhi and Pellegrina (2023), which features a fixed cost for setting up land and two agricultural technologies (traditional and modern). These features allow for responses to changes in trade costs at both the extensive and intensive margins of land-use. On the theoretical side, I generalize the model by incorporating a livestock sector. Based on Gouel and Laborde (2021), livestock is produced using labor and a CES composite of grass and tradable non-forrage crops. This addition is crucial to account for grazing land, which represents 68% of global land-use. To calibrate the model with this new sector, I use data on livestock production and feed consumption from FAOSTAT and GTAP. I also use cost shares data from USDA to determine factor intensity parameters. Empirically, I add to Farrokhi and Pellegrina (2023) by considering a wider range of crops. This provides greater exhaustiveness in terms of land-use.

I then consider the impact of land-use on two aspects of bird biodiversity: species richness and abundance. First, I rely on the Countryside Species-Area model (Pereira and Daily, 2006; Chaudhary et al., 2015; Scherer et al., 2023) to predict long-term *species richness* in 700 ecoregions of the world. The model is able to predict the number of bird species committed to extinction for any regional distribution of land. In particular, it accounts for the fact that the impact of human-modified land on biodiversity is type- and intensity-specific. On average, pastureland is indeed considered less hostile to birds than subsistence farming, and even less so than modern agriculture. Second, I adopt a Verhulst growth model to predict the *abundance* of common bird species within 393 smaller ecoregions of the United States. Crucially, abundance in a given year depends on the ecoregion's agricultural and climate history (Mouysset et al., 2019; Cocco, Kervinio, and Mouysset, 2023). I calibrate the model for the 100 most abundant bird species using count data from the North American Breeding Bird Survey, agricultural land-use and climate variables from ESA, and pesticide use from the US Geological Survey.

In the current version of the paper, I perform the same counterfactual exercises as Farrokhi and Pellegrina (2023): I set back trade costs in agricultural outputs and/or inputs to their 1980 levels. In the baseline and counterfactual scenarios, I (1) simulate the equilibrium in terms of land-use at the 5 arc-minute resolution using the trade component of the model and (2) predict bird biodiversity by feeding the simulated land-use patterns into the ecological components. Preliminary results without the livestock sector indicate that, on average across world ecoregions, species loss would be 9% higher in a counterfactual economy without historical liberalization policies. Distinguishing between liberalization in outputs and in inputs, I find two additional results. First, inputs liberalization appears to have a positive impact on species richness overall: the direct negative effects of modernization on birds survival are more than offset by the benefits associated with the productivity-driven reductions in cropland. Second, outputs liberalization significantly affects the spatial distribution of losses: the reduction in trade costs mostly shifts agricultural activities southwards, thus preventing species loss in the North to the detriment of biodiversity in the South. This is partly offset when also allowing for liberalization in inputs, as it raises yields in the South and constrains the extensive margin of agricultural expansion. Eventually, results on abundance of common birds in the United States suggest that while trade-driven land-use alterations do affect population trends, their impact is species-specific and relatively minor compared with typical year-to-year fluctuations. The next steps are to 1) explore the effects of additional counterfactual policies, (2) decompose the GE effects of trade on biodiversity and (3) modelize the feedback effects of changes in bird biodiversity on agricultural productivity.

Related literature. This paper contributes to three strands of the literature. First, it expands the relatively sparse body of work examining the biodiversity impacts of trade. Much of the work has been confined to quantifying the amount of biodiversity loss *embodied* in bilateral trade links, using input-ouput tables (Chaudhary and Kastner, 2016; Cabernard, Pfister, and Hellweg, 2024). By contrast, my structural approach allows to assess how different the state of biodiversity would be if alternative trade policies were implemented. Therefore, I can measure the extent to which international trade *causes* biodiversity loss. The closest paper related to my work is Polasky, Costello, and McAusland (2004), which provides theoretical results on how trade can affect biodiversity by combining a simple two-country two-good model with a species–area relationship. My contributions with regard to this paper are fourfold: I provide a general equilibrium framework with multiple sectors, multiple countries and incomplete specialization; I move beyond theory by bringing the model to the data; I consider a new margin of agricultural land-use—the intensive margin—; and I adopt an additional biodiversity metric—abundance.

Second, my work contributes to the growing body of literature examining the relationship between agriculture and the environment using quantitative trade models. So far, papers have focused on climate change (Costinot, Donaldson, and Smith, 2014; Gouel and Laborde, 2021), deforestation (Dominguez-Iino, 2025; Farrokhi et al., 2025) and water resources (Carleton, Crews, and Nath, 2025). To my knowledge, this paper is the first to consider biodiversity.

Finally, this paper engages with the emerging literature on the economics of biodiversity and ecosystem services. Earlier work has quantified the benefits that birds provide: Frank and Sudarshan (2024) demonstrated that the functional extinction of Indian vultures increased human mortality by more than 4% and generated roughly \$70 billion in annual social costs, while Noack, Engist, and Larsen (2025) estimated that the 30% decline in bird abundance observed across North America is likely to have reduced U.S. agricultural sales by about 4%. Building on these findings, my contribution is twofold. First, I investigate the drivers behind biodiversity loss. Second, I extend the analysis of economic consequences beyond agricultural productivity to include projected impacts on land-use decisions and international trade, thereby creating a closed feedback loop that captures how changes in biodiversity *shape* and *are shaped* by economic outcomes.

The rest of the paper is organized as follows. Section 2 presents the model. Section 3 details the estimation strategy that allows to take the model to the data. Section 4 presents the results of the counterfactual exercises, and Section 5 concludes.

2 Theory

In this section, I present the general equilibrium model that allows me to link changes in trade costs to changes in bird biodiversity. The model draws upon the works of Farrokhi and Pellegrina (2023) and Gouel and Laborde (2021) for the trade and land-use component, and of Scherer et al. (2023) and Cocco, Kervinio, and Mouysset (2023) for the biodiversity components.

2.1 Environment

I consider a world composed of multiple countries, indexed by $i \in \mathcal{I} \equiv \{1, \ldots, I\}$, and multiple ecoregions, indexed by $e \in \mathcal{E} \equiv \{1, \ldots, E\}$, which delimit distinct ecosystems. A country may contain several ecoregions, and an ecoregion may span several countries. Each country i is endowed with a quantity L_i of land, N_i of labor and V_n of raw fertilizer. Land is divided into heterogeneous fields f. Each field is located at the intersection of some country i, such that $\sum_{f \in \mathcal{F}_i} L_f = L_i$, and some ecoregion e, such that $\sum_{f \in \mathcal{F}_e} L_f = L_e$. All fields correspond to 5-arcminute grid cells and consist of a continuum of heterogeneous plots $\omega \in f$.

Goods are indexed by $g \in \mathcal{G}$ and include: (1) an outside good $g = 0$, (2) agricultural inputs, indexed by $j \in \mathcal{J} \subset \mathcal{G}$, and (3) agricultural outputs, indexed by $k \in \mathcal{K} \subset \mathcal{G}$. As opposed to Farrokhi and Pellegrina (2023), I consider that the bundle of agricultural outputs includes a livestock product, indexed by $k = \ell$, and a forage crop (grass), which is only used to feed livestock and cannot be internationally traded. Similar to Gouel and Laborde (2021), the set of agricultural outputs can therefore be divided into two categories: tradable outputs, indexed by $k \in \mathcal{K}^a$, which include all crops except grass; and crops, indexed by $k \in \mathcal{K}^c$, which include all agricultural outputs except the livestock product.

2.1.1 Preferences

I assume that consumers' preferences follow a three-tier Armington structure. In the upper-tier, the representative household in field f derives utility from consuming the outside good composite, C^0_{fi}, and an agricultural composite, C^a_{fi} according to a CES aggregator:

$$C_{fi} = \left[\left(\beta^0_i\right)^{1/\eta} \left(C^0_{fi}\right)^{(\eta-1)/\eta} + \left(\beta^1_i\right)^{1/\eta} \left(C^a_{fi}\right)^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}, \tag{1}$$

where η is the elasticity of subsitution, β^0_i and β^1_i are exogenous demand shifters. In the middle-tier, the agricultural consumption bundle aggregates the consumption of tradable agricultural outputs $k \in \mathcal{K}^a$ according to CES preferences with elasticity of substitution κ and exogenous demand shifters b^k_i:

$$C^a_{fi} = \left[\sum_{k \in \mathcal{K}^a} \left(\beta^k_i\right)^{1/\kappa} \left(C^k_{fi}\right)^{(\kappa-1)/\kappa} \right]^{\kappa/(\kappa-1)}. \tag{2}$$

Eventually, in the lower-tier, the representative household purchase varieties of each good g from different origin countries according to a CES function with elasticity of substitution σ_g and exogenous demand shifters b^g_{ni}:

$$C^g_{fi} = \left[\sum_{n \in \mathcal{I}} \left(\beta^g_{ni}\right)^{1/\sigma_g} \left(C^g_{nfi}\right)^{(\sigma_g-1)/\sigma_g} \right]^{\sigma_g/(\sigma_g-1)}. \tag{3}$$

2.1.2 Technology

Crop production. Each plot ω is initially covered of natural habitat for birds that is unsuitable for crop production. Following Farrokhi and Pellegrina (2023), I assume that farmers must pay a fixed cost $z^0_{fi}(\omega)$ in units of the outside good to convert their plot for agricultural use. They then choose which crop $k \in \mathcal{K}^c$ to grow and which technology $\tau \in \mathcal{T}$ to use on that plot. By combining land $L^{k\tau}_{fi}(\omega)$, labor $N^{k\tau}_{fi}(\omega)$ and intermediates $M^{k\tau}_{fi}(\omega)$, the farmer of plot ω can produce:

$$Q^{k\tau}_{fi}(\omega) = \left[\frac{1}{\gamma^{k\tau}_L} z^{k\tau}_{fi}(\omega) L^{k\tau}_{fi}(\omega) \right]^{\gamma^{k\tau}_L} \left[\frac{1}{\gamma^{k\tau}_N} N^{k\tau}_{fi}(\omega) \right]^{\gamma^{k\tau}_N} \left[\frac{1}{\gamma^{k\tau}_M} M^{k\tau}_{fi}(\omega) \right]^{\gamma^{k\tau}_M} \tag{4}$$

where $z^{k\tau}_{fi}(\omega)$ is the productivity of land for the crop-technology pair (k, τ) in plot ω. The technology can be either traditional, $\tau = 0$, or modern, $\tau = 1$, and is characterized by intensity parameters $\gamma^{k\tau}_L \in [0, 1]$, $\gamma^{k\tau}_N \in [0, 1]$ and $\gamma^{k\tau}_M = 1 - \gamma^{k\tau}_L - \gamma^{k\tau}_N \in [0, 1]$. Traditional technology is assumed to use only land and labour (i.e. $\gamma^0_M = 0$), whereas modern technology uses all three factors.

The bundle of intermediates $M^{k\tau}_{fi}(\omega)$ is a Cobb-Douglas combination of agricultural inputs j:

$$M^{k\tau}_{fi}(\omega) = \prod_{j\in\mathcal{J}} \left(M^{k\tau j}_{fi}(\omega)\right)^{\gamma^{kj}_M}, \tag{5}$$

where $M^{k\tau j}_{fi}(\omega)$ denotes the use of agricultural input j. The intensity parameters $\gamma^{kj}_M \in [0,1]$ satisfy $\sum_{j\in\mathcal{J}} \gamma^{kj}_M = 1$. Farmers can buy agricultural inputs from different origins according to CES preferences with elasticity of substitution σ_j and exogenous demand shifters b^j_{ni}:

$$M^{k\tau j}_{fi}(\omega) = \left[\sum_{n\in\mathcal{I}} \left(\beta^j_{ni}\right)^{1/\sigma_j} \left(M^{k\tau j}_{nfi}(\omega)\right)^{(\sigma_j-1)/\sigma_j}\right]^{\sigma_j/(\sigma_j-1)}. \tag{6}$$

In each plot, the required fixed cost $z^0_{fi}(\omega)$ and land productivities $z^{k\tau}_{fi}(\omega)$ are drawn from field-specific nested Fréchet distributions introduced by Farrokhi and Pellegrina (2023):

$$\Pr\left(z^0_{fi}(\omega) \le z^0_{fi}\right) = \exp\left\{-\bar{\phi}\left(\frac{z^0_{fi}}{a^0_{fi}}\right)^{-\theta_1}\right\}, \tag{7}$$

$$\Pr\left(z^{k\tau}_{fi}(\omega) \le z^{k\tau}_{fi}\right) = \exp\left\{-\bar{\phi}\left[\sum_{\tau\in\mathcal{T}}\left(\frac{z^{k\tau}_{fi}}{a^{k\tau}_{fi}}\right)^{-\theta_2}\right]^{\theta_1/\theta_2}\right\} \;\forall k \in \mathcal{K}^c, \tag{8}$$

where $\bar{\phi} \equiv [\Gamma(1-1/\theta_1)]^{-\theta_1}$ is a constant set to ensure that $\mathbb{E}[z^0_{fi}] = a^0_{fi}$ and $\mathbb{E}[z^{k\tau}_{fi}] = a^{k\tau}_{fi}$. The parameters θ_1 and θ_2 control the heterogeneity of land productivity draws across crops and across technologies within each crop respectively.

Livestock production. The livestock product $k = \ell$ is produced using labor N^ℓ_i and feed-mix F_i:

$$Q^\ell_i = \left[\frac{1}{\gamma^\ell_N} N^\ell_i\right]^{\gamma^\ell_N} \left[\frac{1}{\gamma^\ell_F} F_i\right]^{\gamma^\ell_F} \tag{9}$$

where $\gamma^\ell_N \in [0,1]$ and $\gamma^\ell_F = 1 - \gamma^\ell_N \in [0,1]$ are intensity parameters.[1] Livestock producers choose which crops to put in the feed-mix F_i according to CES preferences with elasticity of

[1]Contrary to Gouel and Laborde (2021), I do not assume that labor and feed are perfect complements, which simplifies the calibration. They assume that $Q^\ell_i = \min\left(F_i/\mu_i, N^\ell_i/\nu^\ell_i\right)$, where μ_i corresponds to the feed conversion ratio and ν^ℓ_i to the unit labor requirement in country i. In the absence of estimates for the two parameters, I choose to rely on a standard Cobb-Douglas production function. Based on this assumption, I only require data on the cost shares of the two factors.

substitution ζ and exogenous technological shifters $b_i^{k,feed}$:

$$F_i = \left[\sum_{k \in \mathcal{K}^c} \left(\beta_i^{k,feed}\right)^{1/\zeta} \left(F_i^k\right)^{(\zeta-1)/\zeta} \right]^{\zeta/(\zeta-1)} \tag{10}$$

where F_i^k is the demand for crop k in the feed-mix, which on its turn aggregates over all varieties of crop k. As in Gouel and Laborde (2021), I assume that the preferences of livestock producers over crop origins are similar to those of the representative household:

$$F_i^k = \left[\sum_{n \in \mathcal{I}} \left(\beta_{ni}^k\right)^{1/\sigma_k} \left(F_{ni}^k\right)^{(\sigma_k-1)/\sigma_k} \right]^{\sigma_k/(\sigma_k-1)} \quad \forall k \in \mathcal{K}^a. \tag{11}$$

Assuming that grass cannot be traded implies that livestock producers only buy domestic grass, i.e. $F_i^{grass} = F_{ii}^{grass}$.

Other productions. All other sectors work as in Farrokhi and Pellegrina (2023). Processed fertilizers, denoted $v \in \mathcal{J}$, are produced using the domestic supply of raw fertilizers: $Q_i^v = V_i$. The outside good $g = 0$ and nonfertilizer agricultural inputs $j \in \mathcal{J} \setminus v$ are produced under constant returns to scale using labor only: $Q_i^g = A_i^g N_i^g$, with labor productivity A_i^g.

2.1.3 Labor market

Workers are mobile between sectors but immobile between countries. Each worker φ in country i is endowed with efficiency units $\epsilon_i^s(\varphi)$ for each sector s drawn from Fréchet distribution:

$$\Pr(\epsilon_i^s(\varphi) \le \epsilon) = \exp\left\{ -\left[\Gamma(1 - 1/\psi)\right]^{-\psi} \left(\frac{\epsilon}{g_i^s}\right)^{-\psi} \right\} \tag{12}$$

where ψ is between-sector labor supply elasticity, and g_i^s is a scale parameter. Workers can supply their efficiency units either to the non-crop sector ($s = 0$) for wage w_i^0 or to the crop sector ($s = 1$) for wage w_i^1, according to which yields the highest revenue $w_i^s \epsilon_i^s(\varphi)$.

2.1.4 Trade system

Following Farrokhi and Pellegrina (2023), goods are traded across fields via a *hub-and-spoke* system, subject to iceberg trade costs: $d_{fif'n}^g \ge 1$ units of g must be shipped from field f in country i to field f' in country n to sell 1 unit. This trade cost can be decomposed as:

$$d_{fif'n}^g = d_{fi}^g \times d_{in}^g \times d_{f'n}^g \tag{13}$$

where d_{fi}^g is the domestic trade cost between field f and its nearest hub in origin country i, d_{in}^g is the hub-to-hub international trade cost, and $d_{f'n}^g$ is the domestic trade cost between field f' and its nearest hub in destination country n. I make three assumptions regarding domestic trade costs. First, the outside good and the livestock product are not subject to any domestic trade costs, i.e. $d^0_{fif'n} = d^0_{in}$ and $d^\ell_{fif'n} = d^\ell_{in}$. Second, agricultural inputs are not subject to domestic trade costs in the origin country, i.e. $d^j_{fif'n} = d^j_{in} \times d^j_{f'n}$. Third, crops are not subject to domestic trade costs in the destination country, i.e. $d^j_{fif'n} = d^j_{fi} \times d^j_{in}$. As such, everything works as if all goods but crops were produced in hubs, and all goods but agricultural inputs were consumed in hubs. Similarly, w^0_i can be interpreted as a *urban* wage and w^1_i as a *rural* wage.

Goods are not differentiated by fields or hubs. As a result, producer prices including domestic trade costs must be equalized in origin country i, i.e. $p^k_{fi} d^k_{fi} = p^k_i$; producers in more remote fields receive a lower price for crop k. Consumer prices including domestic trade costs must also be equalized in destination country n, i.e. $p^j_{f'n} = p^j_{in} d^j_{in}$; consumers in more remote fields pay a higher price for agricultural input j.

2.1.5 Biodiversity

Land can be categorized as either natural habitat or agricultural.[2] I then decompose *current* land-use in ecoregion e as:

$$L_e = L^{nat}_e + \sum_k \sum_\tau L^{k\tau}_e \tag{14}$$

where $L^{nat}_e = \sum_{f \in \mathcal{F}_e} L^{nat}_{fe}$ is the remaining natural habitat area and $L^{k\tau}_e = \sum_{f \in \mathcal{F}_e} L^{k\tau}_{fe}$ is the area dedicated to the crop-technology pair (k, τ) in ecoregion e.

Species richness. Given this distribution of land, the Countryside Species-Area model (Pereira and Daily, 2006) can be used to estimate a species extinction rate, s^{lost}_e, in every ecoregion e:

$$s^{lost}_e = 1 - \left(\frac{L^{nat}_e + \sum_k \sum_\tau h^{k\tau}_e \times L^{k\tau}_e}{L_e} \right)^{z_e}. \tag{15}$$

Crucially, the relationship accounts for birds' affinity for each type-intensity pair (k, τ) of agricultural land-use through parameters $h^{k\tau}_e \in [0, 1]$. It is assumed that $h^{nat}_e = 1$ —natural habitat can support all bird species—and that affinity decreases as land-use becomes increasingly hostile for birds. For example, it is reasonable to assume that the survival rate of birds is lower in modern agricultural areas than in traditional ones. Prior to any alteration in land-use, I assume that every ecoregion consisted solely of natural habitats and was home to a number S^0_e of bird species. Using Eq. (15), I can predict the number of bird species

[2] I exclude other human-modified land categories, such as urban areas, which are quantitatively negligible relative to agricultural areas.

committed to extinction in the medium- to long-term in ecoregion e:

$$S_e^{lost} = \sum_k \sum_\tau \underbrace{\frac{L_e^{k\tau}\left(1 - h_e^{k\tau}\right)}{\sum_k \sum_\tau \left[L_e^{k\tau}(1 - h_e^{k\tau})\right]} \frac{1}{L_e^{k\tau}} s_e^{lost}}_{CF_e^{k\tau}} S_e^0 L_e^{k\tau} \tag{16}$$

where $CF_e^{k\tau}$, referred to as *characterization factors* in the ecology literature, measure the regional species loss per m^2 of land-use (k, τ).

Abundance. I assume that population follows a Verhulst logistic growth model:

$$A_e^b(t+1) = A_e^b(t)\left[1 + r^b - r^b \frac{A_e^b(t)}{K_e^b(t)}\right] \tag{17}$$

The number of individuals of species b in ecoregion e at time $t+1$, $A_e^b(t+1)$, therefore depends on (1) the population size in the previous year, $A_e^b(t)$; (2) a specific natural growth rate—essentially related to the number of eggs a female can lay without environmental pressure—, r^b; and (3) the carrying capacity of regional ecosystem e for species b in year t, $K_e^b(t)$. The carrying capacity refers to the maximum number of individuals of species b the habitat can sustain, given current ecological conditions. Following Mouysset et al. (2019) and Cocco, Kervinio, and Mouysset (2023), I assume that it is a function of land-use surfaces, $L_e^k(t)$, and climate, $Climate_e(t)$. I consider further that $K_e^b(t)$ directly depends on the total quantity of each agricultural input j used in the ecoregion, $M_e^j(t)$. Thus,

$$\frac{1}{K_{fi}^b(t)} = \iota^b + \sum_{k \in \mathcal{K}^c} \iota_k^b L_e^k(t) + \sum_{j \in \mathcal{J}} \iota_j^b M_e^j(t) + \iota_{\text{clim}}^b Climate_e(t), \tag{18}$$

where ι^b is species fixed effect, ι_k^b assesses the response of species b to land-use k, ι_j^b to the use of input j, and ι_{clim}^b to climate conditions.

2.2 General equilibrium

2.2.1 Good demand

Given equations (1), (2), (3) and (13), utility maximization by the representative household in (f, i) requires

$$C_{fi}^1 = \left(\frac{P_{fi}^a}{P_{fi}}\right)^{-\eta} \beta_i^1 C_{fi}, \tag{19}$$

$$C_{fi}^k = \left(\frac{P_{fi}^k}{P_{fi}^a}\right)^{-\kappa} \beta_i^k C_{fi}^1, \tag{20}$$

and

$$C^g_{nfi} = \left(\frac{p^g_n d^g_{ni} d^g_{fi}}{P^g_{fi}}\right)^{-\sigma_g} \beta^g_{ni} C^g_{fi}. \tag{21}$$

Price indexes that enter the above tiers are:

$$P_{fi} = \left[\beta^0_i (P^0_{fi})^{1-\eta} + \beta^1_i (P^1_{fi})^{1-\eta}\right]^{\frac{1}{1-\eta}}, \tag{22}$$

$$P^1_{fi} = \left[\sum_{k \in \mathcal{K}} \beta^k_i (P^k_{fi})^{1-\kappa}\right]^{\frac{1}{1-\kappa}}, \tag{23}$$

and

$$P^g_{fi} = \left[\sum_{n \in \mathcal{I}} \beta^g_{ni} (p^g_n d^g_{ni} d^g_{fi})^{1-\sigma_g}\right]^{\frac{1}{1-\sigma_g}}. \tag{24}$$

Given assumptions on domestic trade costs, $P^g_{fi} = P^g_i = \left[\sum_{n \in \mathcal{I}} \beta^g_{ni} (p^g_n d^g_{ni})^{1-\sigma_g}\right]^{\frac{1}{1-\sigma_g}}$ for $g = 0$ and $g \in \mathcal{K}^a$, and for all fields $f \in \mathcal{F}_i$.

2.2.2 Production

Crop production. The marginal cost of crop k using technology τ in plot ω of (f, i), $c^{k\tau}_{fi}(\omega)$, is

$$c^{k\tau}_{fi}(\omega) = \left(\frac{r^{k\tau}_{fi}(\omega)}{z^{k\tau}_{fi}(\omega)}\right)^{\gamma^{k\tau}_L} (w^1_i)^{\gamma^{k\tau}_N} (m_{fi})^{\gamma^{k\tau}_M}$$

where $r^{k\tau}_{fi}(\omega)$ is the gross rental price of plot ω and $m_{fi} \equiv \prod_{j \in \mathcal{J}} \left(P^j_{fi}\right)^{\gamma^{kj}_M}$ is the price index of the bundle of agricultural inputs in destination (f, i). Perfect competition in the crop market implies $c^{k\tau}_{fi}(\omega) = p^k_{fi}$. Thus, as shown in Farrokhi and Pellegrina (2023),

$$r^{k\tau}_{fi}(\omega) = z^{k\tau}_{fi}(\omega) h^{k\tau}_{fi} \tag{25}$$

with

$$h^{k\tau}_{fi} \equiv p^k_{fi} \underbrace{\left(\frac{w^1_i}{p^k_{fi}}\right)^{-\gamma^{k\tau}_N/\gamma^{k\tau}_L} \left(\frac{m_{fi}}{p^k_{fi}}\right)^{-\gamma^{k\tau}_M/\gamma^{k\tau}_L}}_{\equiv \tilde{h}^{k\tau}_{fi}}$$

summarizing market prices effects. In equilibrium, the land share of the crop-technology pair (k,τ) in (f,i), $\pi_{fi}^{k\tau}$, is

$$\pi_{fi}^{k\tau} = \alpha_{fi}^{k} \times \alpha_{fi}^{k\tau} \tag{26}$$

where

$$\alpha_{fi}^{k\tau} = \frac{\left(a_{fi}^{k\tau} h_{fi}^{k\tau}\right)^{\theta_2}}{\left(H_{fi}^{k}\right)^{\theta_2}} \tag{27}$$

is the share of land within crop k allocated to technology τ, and

$$\alpha_{fi}^{k} = \frac{\left(H_{fi}^{k}\right)^{\theta_1}}{(a_{fi}^{0} P_{fi}^{0})^{\theta_1} + \sum_{k\in\mathcal{K}^c} \left(H_{fi}^{k}\right)^{\theta_1}} \tag{28}$$

is the share of land allocated to crop k. The term H_{fi}^{k} denotes the aggregate return to crop k:

$$H_{fi}^{k} = \left[\sum_{\tau\in\mathcal{T}} \left(a_{fi}^{k\tau} h_{fi}^{k\tau}\right)^{\theta_2}\right]^{1/\theta_2}. \tag{29}$$

Let $\Omega_{fi}^{k\tau}$ be the set of plots ω in field f to which crop-technology (k,τ) is optimally allocated. Combining Eq. (25)-(29), production of (k,τ) in plot ω is given by

$$Q_{fi}^{k\tau}(\omega) = \begin{cases} \left(\gamma_L^{k\tau}\right)^{-1} \tilde{h}_{fi}^{k\tau} z_{fi}^{k\tau}(\omega), \ \omega \in \Omega_{fi}^{k\tau} \\ 0, \ \omega \notin \Omega_{fi}^{k\tau}, \end{cases} \tag{30}$$

and at the field level by

$$Q_{fi}^{k\tau} = L_{fi} \left(\gamma_L^{k\tau}\right)^{-1} \tilde{h}_{fi}^{k\tau} a_{fi}^{k\tau} \left(\alpha_{fi}^{k}\right)^{(\theta_1-1)/\theta_1} \left(\alpha_{fi}^{k\tau}\right)^{(\theta_2-1)/\theta_2}. \tag{31}$$

Aggregate output of crop k, Q_i^k at the location of hubs in country i is

$$Q_i^k = \sum_{\tau\in\mathcal{T}} Q_i^{k\tau} \tag{32}$$

with

$$Q_i^{k\tau} = \sum_{f\in\mathcal{F}_i} \frac{Q_{fi}^{k\tau}}{d_{fi}^{k}}. \tag{33}$$

Lastly, the quantity of outside good required for setting up land in country i, FC_i^0, equals

$$FC_i^0 = \sum_{f \in \mathcal{F}_i} L_{fi} a_{fi}^0 \left[1 - \left(1 - \sum_{k \in \mathcal{K}^c} \alpha_{fi}^k \right)^{(\theta_1 - 1)/\theta_1} \right]. \tag{34}$$

Livestock production. The share of crop k in the feed-mix of country i is

$$\frac{F_i^k}{F_i} = \beta_i^{k,feed} \left(\frac{P_i^{feed}}{P_i^k} \right)^{-\zeta} \tag{35}$$

with

$$P_i^{feed} = \left[\sum_{k \in \mathcal{K}^c} \beta_i^{k,feed} (P_i^k)^{1-\zeta} \right]^{\frac{1}{1-\zeta}}. \tag{36}$$

Perfect competition in the livestock sector then implies

$$c_i^\ell = \left(w_i^1 \right)^{\gamma_\ell^N} \left(P_i^{feed} \right)^{\gamma_\ell^F} = p_i^\ell. \tag{37}$$

2.2.3 Labor supply

The share of labor supply in sector s, n_i^s, equals

$$n_i^s = \frac{N_i^s}{N_i} = \frac{(g_i^s w_i^s)^\psi}{\sum_{s \in \{0,1\}} (g_i^s w_i^s)^\psi} \tag{38}$$

and aggregate supply of efficiency units to sector s, $\tilde{N}_i^s$, is

$$\tilde{N}_i^s = g_i^s \, (n_i^s)^{(\psi - 1)/\psi} \, N_i. \tag{39}$$

2.2.4 International trade

Total import demand for crop $k \in \mathcal{K}^c$ in country i is the sum of demand for final consumption and for feed-mix

$$\frac{X_i^k}{P_i^k} = C_i^k + F_i^k \tag{40}$$

where X_i^k is the total value of imports. The share of good g from origin n in total imports value of destination (f, i) is

$$\frac{X_{nfi}^g}{X_{fi}^g} = \beta_{ni}^g \left(\frac{p_n^g d_{ni}^g d_{fi}^g}{P_{fi}^g} \right)^{1-\sigma_g}. \tag{41}$$

Given assumptions on internal trade costs, we observe that for $g = 0$ and $g \in \mathcal{K}^a$, $\frac{X^g_{nfi}}{X^g_{fi}} = \frac{X^g_{ni}}{X^g_i}$ for all fields $f \in \mathcal{F}_i$.

2.2.5 Market clearing conditions

The goods market clears for the outside good $g = 0$, agricultural inputs $j \in \mathcal{J}$, grass and other agricultural outputs $k \in \mathcal{K}^a$ when supply in the origin country n equals world demand:

$$p^0_n Q^0_n = \sum_{i \in \mathcal{I}} X^0_{ni} + P^0_n L^0_n, \tag{42}$$

$$p^j_n Q^j_n = \sum_{i \in \mathcal{I}} \sum_{f \in \mathcal{F}_i} \sum_{k \in \mathcal{K}^c} \beta^j_{ni} \left(\frac{P^{feed}_i}{P^{grass}_i} \right)^{-\zeta} \gamma^k_{Mj} \gamma^{k1}_M p^k_i Q^{k1}_{fi}, \tag{43}$$

$$p^{grass}_n Q^{grass}_n = \beta^{grass,feed}_i \left(\frac{P^{feed}_i}{P^{grass}_i} \right)^{-\zeta} \gamma^\ell_F p^\ell_n Q^\ell_n, \tag{44}$$

$$p^k_n Q^k_n = \sum_{i \in \mathcal{I}} \left(\frac{p^k_n d^k_{ni}}{P^k_i} \right)^{-\sigma_k} \beta^k_{ni} \left(\beta^k_i \beta^a_i E_i + \beta^{k,feed}_i \left(\frac{P^{feed}_i}{P^k_i} \right)^{-\zeta} \gamma^\ell_F p^\ell_n Q^\ell_n \right) \quad \forall k \in \mathcal{K}^a. \tag{45}$$

Labor market clearing in every country i requires that labor supply in each sector equal their corresponding labor demand:

$$w^0_i \tilde{N}^0_i = \gamma^\ell_N p^\ell_i Q^\ell_i + p^0_i Q^0_i + \sum_{j \in \mathcal{J}, k \neq v} p^j_i Q^j_i, \tag{46}$$

$$w^1_i \tilde{N}^1_i = \sum_{k \in \mathcal{K}^c} \sum_{\tau \in \mathcal{T}} \gamma^{k\tau}_N p^k_i Q^{k\tau}_i. \tag{47}$$

2.2.6 Expenditures

Total expenditure in country i, E_i, equals the sum of factor rewards

$$E_i = w^0_i \tilde{N}^0_i + w^1_i \tilde{N}^1_i + R_i + p^v_i V_i. \tag{48}$$

where R_i denotes aggregate net land rents in country i.

2.2.7 Definition

Definition 1. For all countries $i, n \in \mathcal{I}$, ecoregions $e \in \mathcal{E}$, fields $f \in \mathcal{F}_i$, goods $g \in \mathcal{G}$ consisting of an outside good, agricultural inputs $j \in \mathcal{J}$, agricultural outputs $k \in \mathcal{K}$, sectors $s \in \{0,1\}$, and technologies $t \in \mathcal{T}$, a global ecnomy is characterized by endowments $\mathcal{D} \equiv \{L_{fi}, N_i, V_i, L_e, S_e^0\}$, supply parameters $\Omega_S \equiv \{\theta_1, \theta_2, \varphi, g_i^s, \gamma_L^{k\tau}, \gamma_M^{k\tau}, \gamma_N^{k\tau}, \gamma_{Mj}^k, a_{fi}^0, a_{fi}^{k\tau}\}$, and demand parameters $\Omega_D \equiv \{\eta, \kappa, \sigma_g, \beta_i^0, \beta_i^1, \beta_i^k, \beta_i^{k,feed}, \beta_{ni}^k, d_{ni}^k, d_{fi}^k, A_i^k\}$.

Definition 2. Given a global economy characterized by $\{\mathcal{D}, \Omega_S, \Omega_D\}$ a general equilibrium consists of prices $\{p_i^g\}$ in all countries $i \in \mathcal{I}$ and for all goods $g \in \mathcal{G}$, such that equations (19)-(48) hold.

3 Estimation

In order to simulate the model described in the previous section, data on endowments, as well as estimates of the demand, supply and ecological parameters, are required. In this section, I present the data used and the methods employed to estimate the parameters missing from the literature. Eventually, I evaluate the model's fit.

3.1 Data

The model is calibrated at three spatial scales—countries, ecoregions, and individual fields. The species-richness component is calibrated across global ecoregions, while the abundance component uses finer-resolution U.S. Level-IV ecoregions. The dataset includes 66 countries (with one representative for the rest of the world), 698 world ecoregions, 393 U.S. Level-IV ecoregions, and 822,053 fields. It covers 36 agricultural outputs (including a livestock aggregate) and three agricultural inputs: pesticides, fertilizers, and farm machinery. I choose 2011 as the baseline year.

Country-level data. For the outside good, agricultural inputs and the agricultural output sector as a whole, I use data on bilateral trade flows and gross outputs built by Farrokhi and Pellegrina (2023) using information from BACI-CEPII, FAOSTAT, WIOD, OECD IOTs, STAN and UNIDO. I also collect data on bilateral trade flows, production weight, production value, land-use, and producer prices for each agricultural output (but grass) from FAOSTAT. Data for the livestock good are derived by aggregating data on meat, eggs, milk and dairy products. The demand of a crop for feed is calculated using FAOSTAT Food Balance Sheets. Eventually, the value of grass production and grass price are computed by combining FAOSTAT data on land-use with data on the value added accruing to land for "Bovine cattle, sheep and goats, horses", "Animal products nec" and "Raw milk" from the GTAP database version 9.2.

Global ecoregion-level data. Global ecoregions are defined by WWF as "large unit of land or water containing a geographically distinct assemblage of species, natural communities, and environmental conditions". I recover the boundaries of each ecoregion from Olson et al. (2001) and the original number of bird species, S_e^0, from Chaudhary and Brooks (2018). My field selection covers on average 66% of an ecoregion area. The original number of birds species varies from 24 to 847, with significant spatial disparities between the northern and the southern hemispheres. I rely on Scherer et al. (2023) for characterization factors, $CF_e^{k\tau}$. Among available type-intensity pairs of land-use, I select the following: cropland minimal, cropland intensive, plantation minimal, plantation intensive, and pasture minimal. I assume that the 'minimal' intensity corresponds to the traditional technology $\tau = 0$ in my model, and 'intensive' to the modern technology $\tau = 1$. I assume further that all crops but palm and grass belong to the type 'cropland', while palm belongs to the type 'plantation' and grass corresponds to 'pasture'.

US ecoregion-level data. To calibrate the population-dynamics model I require fine-scale data on bird abundance, land-use practices, input applications, and climate. Such high-resolution datasets are uniquely available for the United States. For bird abundance I rely on the North American Breeding Bird Survey (NABBS), a long-term, continent-wide monitoring program (Pardieck et al., 2020). NABBS follows a strict, repeatable protocol that yields high-quality trend data: each year, trained volunteers drive predetermined 40km routes, stopping at 50 points per route. At each stop they record all birds seen or heard within a 400m radius. This standardized effort provides consistent, comparable counts across years and regions. Data are available at the species-route-segment-year combination, covering more than 400 bird species on thousands of routes (see Figure A.1 for a map). I select the 100 most abundant bird species in 1992. Land-use information comes from an ESA satellite product that classifies every $0.002° \times 0.002°$ pixel, notably distinguishing between cropland and pasture (cropland pixels are depicted in Figure A.1). To account for local climate, I then collect data on monthly precipitation and temperature from an ESA reanalysis provided on a $0.25° \times 0.025°$ grid, which I aggregate at the seasonal level. Eventually, I use pesticide-application totals from the U.S. Geological Survey reported at the county level. All of these variables are aggregated to the Level-IV ecoregion level by intersecting each original spatial unit with ecoregion boundaries and applying the same area-based weight—the proportion of the unit's area that lies within the ecoregion—to either sum (for bird counts, cropland and pasture area) or average (for climate variables and pesticide use). Overall, the resulting sample spans the years 1992 through 2017.

Field-level data. Potential yields are sourced from the Global Agro-Ecological Zoning (GAEZ) project (Fischer et al., 2021). GAEZ agronomic model produces field-level yield estimates by evaluating each field's edaphic and climatic attributes—including dominant soil type, altitude, slope, temperature, precipitation, and related variables—against the specific requirements of individual crops. In this framework a 'field' corresponds to a 5'×5' grid cell, which approximates an area of 10km×10km. As is common in the literature, I select the estimates for the reference period 1961-1990 under rain-fed conditions. GAEZ also provides yields for different levels of input use. Following Farrokhi and Pellegrina (2023), I map these input regimes onto the technology parameter τ in my model: the low-input regime, representing conventional, subsistence-oriented land management, corresponds to $\tau = 0$ (traditional); the high-input regime, reflecting intensive, market-oriented management with fertilizer, irrigation, and mechanization, corresponds to $\tau = 1$ (modern). To anchor the model in observed land-use patterns, I also use data on the proportions of each field occupied by cropland and pasture circa 2000 from Earthstat (Ramankutty et al., 2008). These figures are derived from a synthesis of national agricultural inventories and satellite-derived land-cover maps.

3.2 Demand-side parameters

The demand-side parameters are estimated directly from the model's micro-foundations, closely following Farrokhi and Pellegrina (2023), and are summarized in Table 1.

Lower-tier demand for agricultural outputs. Based on Eq. (41), I first estimate the elasticity of substitution between crop-varieties σ_k using:

$$\log\left(\frac{X_{ni}^k}{X_i^k}\right) = \delta_i^k + (1-\sigma_k)\log p_n^k + \epsilon_{ni}^k$$

where $\delta_i^k \equiv -\log\left[\sum_{n\in\mathcal{I}} \beta_{ni}^k (p_n^k d_{ni}^k)^{1-\sigma_k}\right]$ is an importer-crop fixed effect and $\epsilon_{ni}^k \equiv \log\left[\beta_{ni}^k \left(d_{ni}^k\right)^{1-\sigma_k}\right]$ is a residual. Setting $\sum_{n\in\mathcal{I}} \epsilon_{ni}^k = 0$, I estimate $\beta_{ni}^k \left(d_{ni}^k\right)^{1-\sigma_k}$ and a different elasticity of substitution σ_k for cereals, perishables and non-perishables. To account for endogeneity, I compute an average of GAEZ potential yields across fields in the exporting country and use it to intrument $\log p_n^k$.

Lower-tier demand for other goods. As in Farrokhi and Pellegrina (2023), I set $\sigma_g = 4$ for the manufacturing good and for agricultural inputs. For $g = \{\text{manufacturing, pesticides, machinery}\}$, I estimate:

$$\log\left(\frac{X_{ni}^g}{X_i^g}\right) - (1-\sigma_g)\log w_n = \delta_i^g + \delta_n^g + \epsilon_{ni}^g$$

where $\delta_i^g \equiv (1-\sigma_g)\log P_i^g$ is a destination fixed effect, $\delta_n^g \equiv (1-\sigma_g)\log A_n^g$ is an origin fixed effect and $\epsilon_{ni}^g \equiv \log\left[\beta_{ni}^g \left(d_{ni}^g\right)^{1-\sigma_g}\right]$. This allows to recover $\beta_{ni}^g \left(d_{ni}^g\right)^{1-\sigma_g}$ and A_n^g. For

$g = \{\text{fertilizers}\}$, I estimate:

$$\log\left(\frac{X^g_{ni}}{X^g_i}\right) - (1-\sigma_g)\log p^g_i = \delta^g_i + \epsilon^g_{ni}$$

and recover $\beta^g_{ni}\,(d^g_{ni})^{1-\sigma_g}$ from residuals.

Middle-tier demand for agricultural outputs. Based on Eq. (20), I then estimate the elasticity of substitution between agricultural outputs κ using:

$$\log\left(\frac{X^k_i}{X^1_i}\right) = \delta_i + (1-\kappa)\log P^k_i + \epsilon^k_i$$

where $\delta_i \equiv (1-\kappa)\log P^1_i$ is a country fixed effect and $\epsilon^k_i \equiv \log\beta^k_i$ is a residual. I assume, without loss of generality, that $\sum_{k\in\mathcal{K}^a}\epsilon^k_i = 0$. P^k_i is constructed using the estimates of σ_k and $\beta^k_{ni}\left(d^k_{ni}\right)^{1-\sigma_k}$ from the previous step following Eq. (24). I address potential endogenity issues by instrumenting $\log P^k_i$ using the average of GAEZ potentiel yield acros fields for crop k in country i. The procedure also allows to recover β^k_i from the residuals.

Upper-tier demand. I borrow from Comin, Lashkari, and Mestieri (2021) and set the elasticity of substitution between agricultural and non-agricultural goods to $\eta = 0.5$. Given this estimate, I recover β^0_i and β^1_i based on Eq. (19) and Eq. (22).

Middle-tier demand for feed. I set the elasticity of substitution between crops in the feed mix to $\zeta = 0.9$, as in Gouel and Laborde (2021). Based on Eq. (35), I then estimate feed demand shifters $\beta^{k,feed}_i$ using:

$$\log F^k_i + \zeta P^k_i = \delta^{feed}_i + \epsilon^{k,feed}_i$$

where $\delta^{feed}_i \equiv \log F_i - \zeta\log P^{feed}_i$ is a country fixed effect and $\epsilon^{feed,k}_i \equiv \log\beta^{k,feed}_i$ is a residual.

3.3 Supply-side parameters

I adopt most of the supply-side parameters from Farrokhi and Pellegrina (2023). The factor- and input-share values are taken as constant across all crops (with the exception of grass in my case, which uses only land). Accordingly, I set: $\gamma^{k1}_L = 0.206$, $\gamma^{k1}_N = 0.207$, $\gamma^{k1}_M = 0.587$ for the modern technology; $\gamma^{k0}_L = 0.551$, $\gamma^{k0}_N = 0.449$ and $\gamma^{k0}_M = 0$ for the traditional technology; $\gamma^{fertilizers}_M = 0.256$, $\gamma^{pesticides}_M = 0.158$ and $\gamma^{machinery}_M = 0.585$ for agricultural input shares within the modern technology. I also set the elasticity of production choices between crops to $\theta_1 = 1.38$ and the elasticity of production choices between technologies to $\theta_2 = 4.51$. Crop-technology productivity shifters are recovered using GAEZ data on potential yields, following the methdology described in Farrokhi and Pellegrina (2023). For livestock

Table 1: Demand-side parameters

Parameters	Description	Source	Estimate
σ_k for cereals	Elast. of subst. between varieties	Estimated from gravity eq.	9.19 (3.78)
σ_k for perishables	Elast. of subst. between varieties	Estimated from gravity eq.	2.96 (0.74)
σ_k for non-perishables	Elast. of subst. between varieties	Estimated from gravity eq.	5.87 (0.56)
σ_g for $g = 0$, $g \in \mathcal{J}$	Elast. of subst. between varieties	Literature	4
κ	Elast. of subst. between ag. outputs	Estimated from gravity eq.	1.25 (0.11)
η	Elast. of subst. between ag. and non-ag.	Comin, Lashkari, and Mestieri (2021)	0.5
$b_{ni}^k (d_{ni}^k)^{1-\sigma_k}$, b_i^k	Demand shifters of ag. outputs	Residuals from gravity eq.	–
$b_{ni}^g (d_{ni}^g)^{1-\sigma_g}$, $g \neq k$	Demand shifters of other goods	Farrokhi and Pellegrina (2023)	–
b_i^0, b_i^1	Demand shifters of sectors	Farrokhi and Pellegrina (2023)	–
d_{fi}^g	Domestic trade costs	Farrokhi and Pellegrina (2023)	–
$b_i^{k,feed}$	Demand shifters of crops in feed	Residuals from FE reg.	–

Notes: The table presents sources and estimation methods for the structural demand-side parameters of the model. Standard errors are clustered at the country of origin and good level.

production, labor- and feed-share parameters are calibrated to USDA-reported cost shares, i.e. $\gamma_N^\ell = 0.38$ and $\gamma_F^\ell = 0.62$, while the elasticity of substitution among crops in the feed mix borrow from the literature review of Gouel and Laborde (2021), i.e. $\zeta = 0.9$. The source and value (when applicable) of all supply-side parameters is reported in Table 2.

3.4 Ecological parameters

As mentioned earlier, the ecological model for species richness is calibrated using characterization factors taken from Scherer et al. (2023). In contrast, the parameters required for the abundance component are absent from the existing literature. Consequently, I calibrate this part of the model with empirical data on bird abundance, land-use patterns, climate variables, and pesticide application across the United States. Combining Eq. (17) and (18), and adjusting for data limitations, I estimate the following equation for each of the 100 bird species

Table 2: Supply-side parameters

Parameters	Description	Source	Value
θ_1	Productivity dispersion betw. crops	Farrokhi and Pellegrina (2023)	1.38
θ_2	Productivity dispersion betw. technologies	Farrokhi and Pellegrina (2023)	4.51
ζ	Elast. of subst. betw. crops in feed	Gouel and Laborde (2021)	0.9
$a_{fi}^{k\tau}$	Crop-technology productivity shifter	Potential yields from FAO-GAEZ	–
a_{fi}^{0}	Investment intensity parameter	Farmland shares from EarthStat	–
A_i^0, A_i^j	Productivity shifters of non-crop goods	Farrokhi and Pellegrina (2023)	–
$\gamma_L^{k\tau}$, $\gamma_N^{k\tau}$, $\gamma_M^{k\tau}$	Crop factor shares	Farrokhi and Pellegrina (2023)	–
γ_M^{kj}	Crop input shares	Farrokhi and Pellegrina (2023)	–
γ_N^{ℓ}, γ_F^{ℓ}	Livestock factor shares	Cost shares from USDA	–
ψ	Between-sector labor supply elasticity	Literature	2

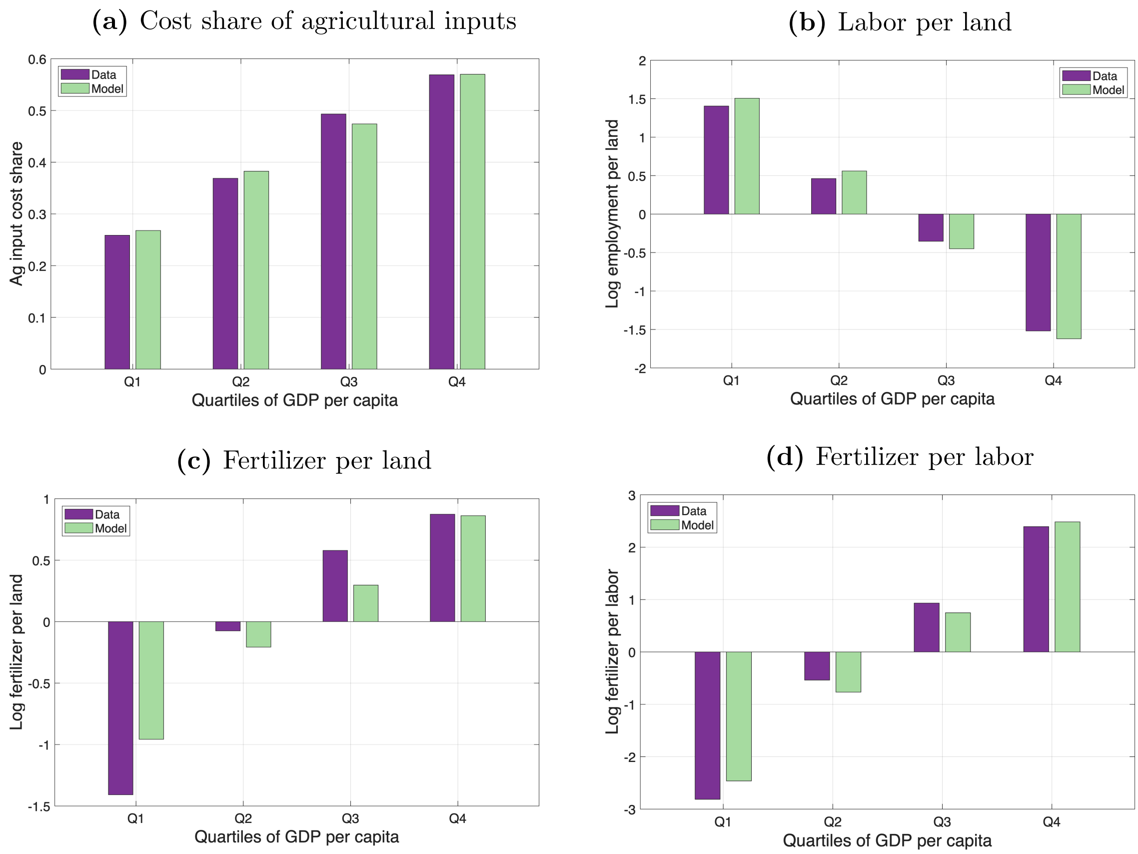
Notes: The table presents sources for the structural supply-side parameters of the model.

in my sample:

$$
\begin{aligned}
A_e^b(t+1) = \alpha_e &+ \beta_s^1 \times A_e^b(t) + \beta_b^2 \times A_e^b(t)^2 + \beta_b^3 \times \text{CROPLAND}_e(t) \times A_e^b(t)^2 \\
&+ \beta_b^4 \times \text{PASTURE}_e(t) \times A_e^b(t)^2 + \beta_b^5 \times \text{PESTICIDES}_e(t) \times A_e^b(t)^2 \\
&+ \beta_b^6 \times \text{TEMP-WINTER}_e(t) \times A_e^b(t)^2 + \beta_b^7 \times \text{TEMP-SPRING}_e(t) \times A_e^b(t)^2 \\
&+ \beta_b^8 \times \text{TEMP-SUMMER}_e(t) \times A_e^b(t)^2 + \beta_b^9 \times \text{TEMP-FALL}_e(t) \times A_e^b(t)^2 \\
&+ \beta_b^{10} \times \text{PRECIP-WINTER}_e(t) \times A_e^b(t)^2 + \beta_b^{11} \times \text{PRECIP-SPRING}_e(t) \times A_e^b(t)^2 \\
&+ \beta_b^{12} \times \text{PRECIP-SUMMER}_e(t) \times A_e^b(t)^2 + \beta_b^{13} \times \text{PRECIP-FALL}_e(t) \times A_e^b(t)^2 \\
&+ \epsilon_e^b
\end{aligned}
\tag{49}
$$

where α_e is a Level-IV ecoregion fixed effects, $\text{CROPLAND}_e(t)$ (resp. $\text{PASTURE}_e(t)$) is the total cropland (resp. pasture) area in ecoregion e in year t, $\text{TEMP-WINTER}_e(t)$ is the mean temperature in ecoregion e during the first quarter of year t—similarly spring corresponds to the second quarter, fall to the third and winter to the fourth–, $\text{PRECIP-WINTER}_e(t)$ corresponds to total precipitation in ecoregion e during the first quarter—with analogous variables for the other seasons— and ϵ_e^b is a residual. The β_b parameters are the species-specific coefficients to be estimated, exploiting within-ecoregion variation in the covariates. I do not incorporate crop-specific effects or other agricultural inputs because the requisite data are

Figure 1: Model fit on measures of input intensity



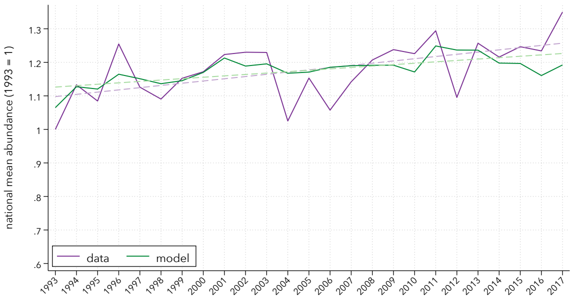
Notes: Each panel in the above figure plots observed (purple bars) and model-predicted (green bars) measures of input intensity for each quartile of GDP per capita. GDP per capita, labor per land, fertilizer per land and fertilizer per labor are normalized according to their global averages.

unavailable. Similarly, I refrain from modelling technological influences beyond the pesticide effect; U.S. agriculture is already largely modernized, making it difficult to disentangle any additional impact of traditional versus modern practices. In the current version of the paper, the parameters are obtained by fitting the model with ordinary least-squares (OLS) regression.

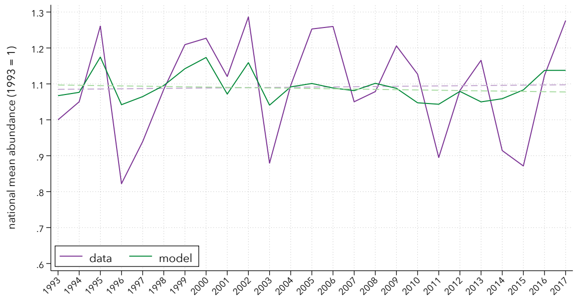
3.5 Other parameters

I consider $d^g_{fi} = d_{fi}$ and take the values of domestic trade costs from Farrokhi and Pellegrina (2023), which were estimated using the methodology of Allen and Arkolakis (2014) and data on the transportation networks of all countries. Following Farrokhi and Pellegrina (2023) again, I set the labor supply elasticity to 2, i.e. $\psi = 2$, and normalize the shifters of labor supply to 1, i.e. $g^1_i = g^0_i = 1$.

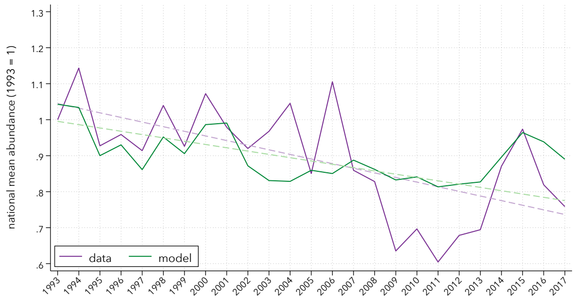
Figure 2: Model fit on national mean abundances of three exemplar species (1993-2017)



(a) Tufted titmouse



(b) Eastern phoebe



(c) Black billed magpie

Notes: Each panel of the above figure plots observed (purple line) and model-predicted (green line) national mean abundances for three exemplar bird species. Dotted lines correspond to linear fits. National mean abundance are normalized by the 1993 historical value.

3.6 Goodness of fit

Land-use. In the current version of this paper, I calibrate the model for 34 crops only, omitting the livestock sector and grass. Following the methodology of Farrokhi and Pellegrina (2023), I evaluate whether the calibrated model reproduces untargeted moments that capture cross-country variation in agricultural input intensity. As illustrated in Figure 1, the model provides an excellent fit to the empirical distribution of several key indicators—cost share of agricultural inputs, labor per unit of land, fertilizer per unit of land, and fertilizer per unit of labor—across all quartiles of GDP per capita. I go further and verify that the calibration generates consistent acreage elasticities, which can be computed as:

$$\frac{dL_i^k}{dp_i^k}\frac{p_i^k}{L_i^k} = \sum_{\tau}\sum_{f}\frac{L_{fi}^{k\tau}}{L_i^k}\frac{1}{\gamma_L^{k\tau}}\left[\theta_1\alpha_{fi}^{k\tau}(1-\alpha_{fi}^k)+\theta_2(1-\alpha_{fi}^{k\tau})\right].$$

Focusing on the United States, the estimated acreage-elasticity for maize is 0.30. This estimate falls within the range of micro-level findings reported for the country—generally between 0.01 and 0.95—as summarized in a brief literature review in Gouel and Laborde (2021).

Abundance. The adjusted R^2 from estimating Eq. (49) for all bird species over the 1992-2017 period ranges from 0.59 to 0.97, with a mean of 0.90. To obtain model-based population forecasts, I iterate the growth equation forward in time. Starting from the observed counts in 1992, the OLS-estimated coefficients are applied year-by-year to project the population of each species in each ecoregion up to 2017. Figure 2 plots the time series of observed and model-predicted national mean abundances for three exemplar species. Each species belongs to a different taxonomic family, occupies a distinct dietary niche, and follows a unique national trend, highlighting the model's ability to capture diverse ecological patterns.

4 Counterfactuals

In this section, I outline the proposed counterfactual exercises and report their results.

4.1 Methodology

I evaluate how agricultural trade policies affect biodiversity by implementing a set of counterfactual simulations, following the approach of Farrokhi and Pellegrina (2023): all model parameters are held at their 2011 values, while the trade costs for both agricultural inputs and outputs are reset to the 1980 levels. This generates alternative land-use patterns, which are then fed into the ecological sub-models. The resulting species-richness and abundance outcomes are compared with the baseline 2011 scenario, allowing me to isolate the net impact of reduced trade barriers on biodiversity. To disentangle the separate contributions, I run two additional scenarios. In the first, only the input trade costs are set to their 1980 levels while output costs remain at 2011 values, isolating the effect of input liberalization. In the second, only the output trade costs are set to 1980 levels while input costs stay at 2011 levels, revealing the impact of output liberalization alone. By contrasting these three outcomes with the baseline, I can identify how changes in trade costs—both jointly and individually—shape bird biodiversity outcomes. In future work I plan to explore additional counterfactual scenarios, such as a global economy in which all agricultural trade is eliminated.

4.2 Results

Species richness. Table 3 summarizes the average species-richness and global land-use outcomes. Column 1 reflects the baseline economy, where trade costs for both agricultural inputs and outputs are set at their 2011 levels. In that scenario the average number of bird species commited to extinction amounts to 4.33 species per ecoregion. Of these losses, 3.15 species are linked to modern, input-intensive agriculture, while 1.17 species stem from traditional, low-intensity farming. Although agricultural land covers only a small share of the total global land surface (0.07% in the baseline), about 72% of agricultural land is classified as modern. The rest of the table shows that reverting trade costs to their 1980 levels would worsen avian

Table 3: Impacts of setting trade costs to their 1980 level on species loss and land-use

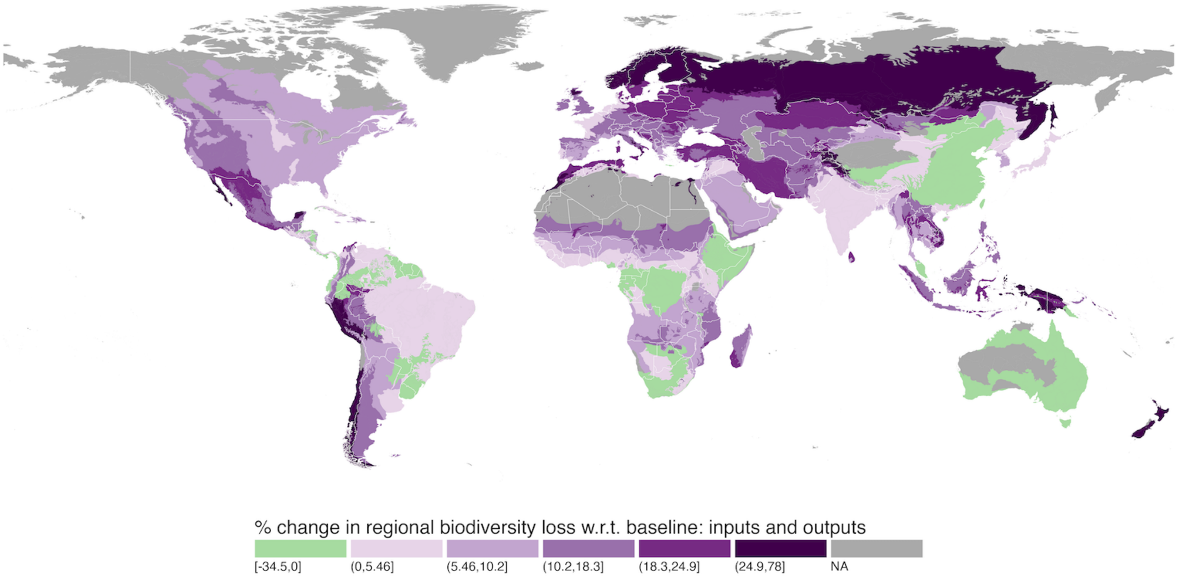
		Changes in trade costs		
	Baseline	Inputs + outputs	Inputs	Outputs
	(1)	(2)	(3)	(4)
Regional species loss (av.)	4.33	+9.25%	+5.83%	+2.93%
due to modern ag. (av.)	3.15	+2.15%	-2.25%	+4.31%
due to trad. ag. (av.)	1.17	+22.48%	+19.99%	+1.91%
Share of ag. land (tot.)	0.07	+10.43%	+7.33%	+2.48%
Share of ag. land in modern (tot.)	0.72	-3.79%	-5.23%	+1.56%

Notes: The table reports the baseline value and the percentage changes of listed variables in the counterfactual with trade costs of agricultural inputs and/or agricultural outputs in 1980 relative to the baseline equilibrium of 2011. Preliminary results without considering the livestock sector and grass production in the estimation of the baseline and counterfactual equilibria.

species losses. Under the full-deliberalization scenario (inputs and outputs) the average regional species loss rises by 9.25% relative to the 2011 baseline. The extra loss is driven largely by a significant increase of land-use at the extensive margin exceeding 10%. Although the proportion of modern agricultural land falls slightly (-3.79% overall), the extensive-margin expansion more than offsets any benefit from reduced exposure to agrochemicals, leading to a net increase in habitat conversion and higher extinction risk. Losses attributable to traditional agriculture surge by 22.5% in this counterfactual, while the impact of modern agriculture changes only modestly although higher in the baseline. These findings imply that trade liberalization, such as the one which occured between 1980 and 2011, has the potential to limit bird-species loss, primarily by curbing the spread of extensive, low-intensity farming. It is worth noting, however, that the gains from trade liberalization are partly offset by the intensification of modern agriculture.

Columns 3 and 4 isolate the effects of altering trade costs for agricultural inputs and outputs respectively. When only the input side becomes more costly (Column 3), the total share of land allocated to agriculture expands by over 7%. At the same time, the proportion of that land classified as modern falls by about 5%, indicating a shift toward more traditional farming practices. This reallocation has two contrasting consequences on biodiversity. The modest decline in the species-loss component associated with modern agriculture indicates that the retreat of intensive farming marginally alleviates pressure on ecosystems. In stark contrast, the species-loss component associated with traditional agriculture rises by more than 22%, driven by small-holder farms expanding cultivation into previously undisturbed habitats. When these opposing effects are aggregated, the net outcome is a deterioration in biodiversity:

Figure 3: Spatial distribution of changes in regional species loss with respect to baseline



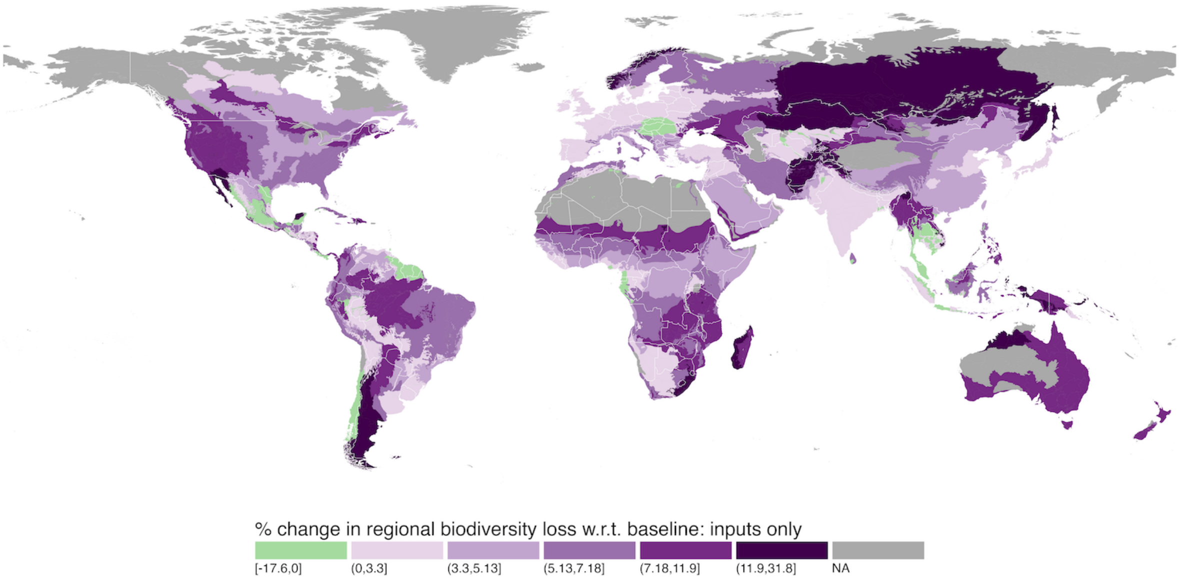
Notes: The figure plots the percentage changes in regional species loss in the counterfactual with trade costs of agricultural inputs and outputs in 1980 relative to the baseline equilibrium of 2011, for each ecoregion. A negative value (in green) indicates that bird species richness is higher in the counterfactual scenario than in the baseline. Preliminary results without considering the livestock sector and grass production in the estimation of the baseline and counterfactual equilibria.

the overall number of species projected to face extinction rises by close to 6%. Hence, the escalation of input trade costs principally drives land-use expansion, overwhelming the modest biodiversity gains derived from a reduced footprint of modern agriculture. Increasing trade costs on the output side (Column 4) generates a qualitatively different response. Higher export barriers diminish the profitability of specialization, compelling nations to produce more for domestic consumption, including crops for which they lack comparative advantage. Because the agro-ecological conditions for these crops are sub-optimal, producers must apply greater quantities of inputs to attain acceptable yields. Pressures at both the extensive and intensive margins produce an increase in average regional species loss of roughly 3%. However, the output-cost channel appears less of a driver of biodiversity change than the input-cost channel.

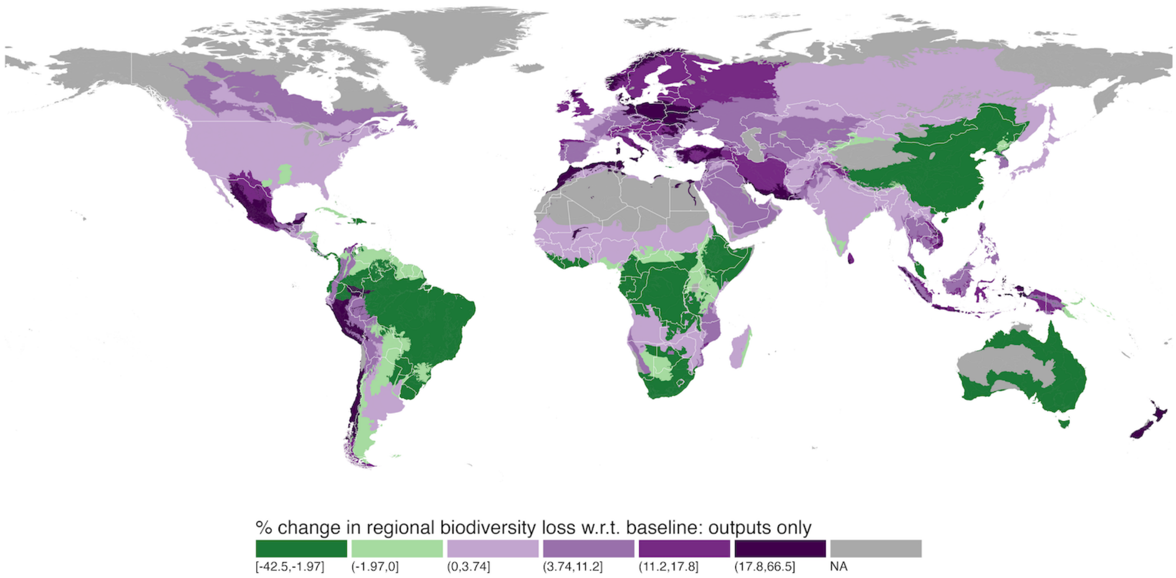
Figure 3 shows the spatial distribution of biodiversity gains and losses in the counterfactual economy. It confirms that raising trade costs for both outputs and inputs generally leads to greater biodiversity loss across most ecoregions. However, the average outcomes reported earlier mask substantial spatial heterogeneity. In several ecoregions, the number of bird species driven toward extinction rises by as much as 78% when trade costs revert to their 1980 levels, underscoring the sizable benefit of trade liberalization in averting losses. Conversely, many southern ecoregions experience improved biodiversity under deliberalization, with the number of threatened species falling by up to 34.5%. As illustrated in Appendix Figure A.2, the regions that suffer under liberalization are precisely those with the highest pre-anthropocene

Figure 4: Decomposition of the impact of setting trade costs to their 1980 level on regional species loss by good type

(a) Only agricultural inputs

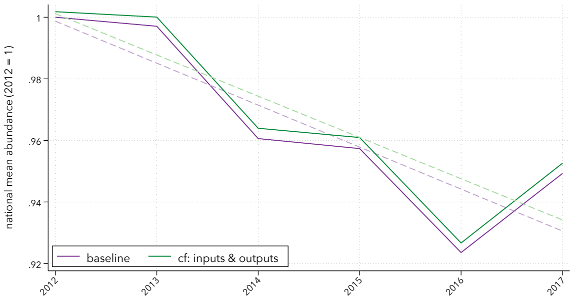


(b) Only agricultural outputs

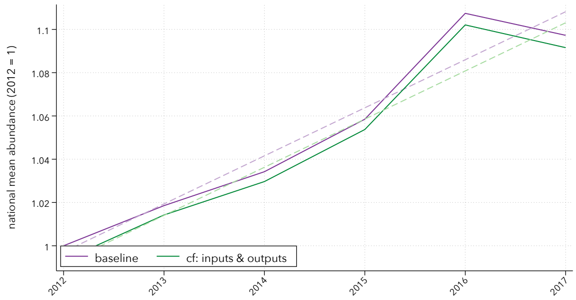


Notes: The figure plots the percentage changes in regional species loss in the counterfactual with trade costs of agricultural inputs (panel a) or agricultural outputs (panel b) in 1980 relative to the baseline equilibrium of 2011, for each ecoregion. A negative value (in green) indicates that bird species richness is higher in the counterfactual scenario than in the baseline. Preliminary results without considering the livestock sector and grass production in the estimation of the baseline and counterfactual equilibria.

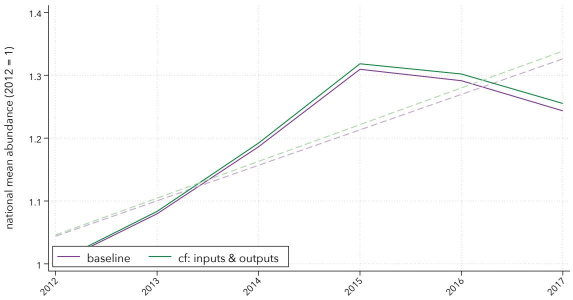
Figure 5: Impacts of setting trade costs to their 1980 level on the population dynamics of three exemplar species (2012-2017)



(a) Tufted titmouse



(b) Eastern phoebe



(c) Black billed magpie

Notes: Each panel of the above figure plots baseline (purple line) and counterfactual (green line) national mean abundances for three exemplar bird species. In the counterfactual scenario, both input and output trade costs are set to their 1980 level. Dotted lines correspond to linear fits. National mean abundance are normalized by the 2012 historical value. Preliminary results without considering the livestock sector and grass production in the estimation of the baseline and counterfactual equilibria.

species richness. These findings suggest that agricultural trade liberalization policies can curb global biodiversity loss overall, but at the cost of key southern biodiversity hotspots. Disentangling the effects of input-cost and output-cost adjustments clarifies the mechanisms at work. As shown in Figure 4, liberalizing outputs while keeping trade in inputs costly increases biodiversity loss in the affected regions relative to the baseline. The reverse pattern emerges when inputs are liberalized but output trade costs remain high: biodiversity improves in these hotspots. Consequently, liberalizing agricultural output markets tends to push production southward into biodiversity-rich areas. Although reductions in inputs trade costs can partially offset habitat conversion by boosting productivity, the first effect dominates, leading to net biodiversity loss.

Abundance. Figure 5 displays the medium-run dynamics of three bird species under two conditions: the baseline and a counterfactual scenario in which both output and input trade costs are set to their 1980 levels. In the counterfactual exercise, land-use is shocked once—

in 2011—and the population-dynamics model projects abundances for the ensuing years.The findings suggest that trade policies can influence the population trajectories of common U.S. birds, but the effects are modest and heterogeneous. Among the three exemplar species, the national-mean abundance of the tufted titmouse and the black-billed magpie would be slightly higher in the counterfactual scenario than in the baseline—by 0.3 percentage points (p.p.) and 1.2 p.p. at the end of the period, respectively—while the eastern phoebe's national-mean abundance would be 0.6 p.p. lower. Although the gap persists over time, it remains small relative to typical year-to-year fluctuations. One limitation of the model is that it aggregates pesticide use only by total quantity, ignoring differences among active ingredients. Recent agro-chemical innovations may have introduced more hazardous compounds that achieve comparable pest control with lower application rates (t/ha), potentially causing the model to underestimate intensive-margin changes and associated biodiversity loss. Another shortcoming is the lack of accounting for alterations in land-use and climate within ecoregions where these birds migrate during winter; this caveat will be addressed in forthcoming versions of the paper.

Feedback effects on agriculture, trade and welfare. To be completed.

5 Conclusion

This paper uses a structural general equilibrium model that connects agricultural trade policies with bird biodiversity outcomes through the land-use channel, considering changes at both the extensive margin—natural habitat destruction—and the intensive margin—ecosystem disruptions associated with the use of agricultural inputs. By linking a multi-country trade model to a countryside species-area framework and a Verhulst population-dynamics model, I quantify how historic reductions in trade costs between 1980 and 2011 may have reshaped global bird species richness and abundance of common birds in the U.S.

Preliminary results show that the overall effect of trade liberalization is a modest reduction in average regional bird species loss. This net benefit for biodiversity arises because productivity gains associated with lower trade barriers reduce the total land area needed for food production, outweighing the negative impacts of intensified farming. However, the aggregate gain conceals strong spatial heterogeneity: southern, biodiversity-rich ecoregions tend to lose more species under liberalization because lower trade costs for agricultural outputs pull production southward, whereas many northern regions experience modest improvements. Eventually, predicted changes in national mea, abundance for three exemplar U.S. bird species are small and species-specific, indicating that trade-induced land-use shifts can influence population trajectories but remain minor relative to normal inter-annual fluctuations.

This paper has several limitations. It aggregates pesticide use without distinguishing among active ingredients, potentially understating the impact of newer, more toxic chemicals.

For the moment, livestock and grass production are omitted from the quantitative exercise, which may bias estimates of biodiversity effects. Additionally, the framework does not incorporate migratory pathways, limiting its ability to capture full-year habitat dependencies for migratory species. Future work will thus focus on exploring additional counterfactuals, extend the ecological component to include migratory connectivity and model the feedback effects of changes in biodiversity on agricultural productivity and welfare.

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A Appendix

A.1 Additional figures

Figure A.1: Distribution of NABBS routes and cropland across the United States

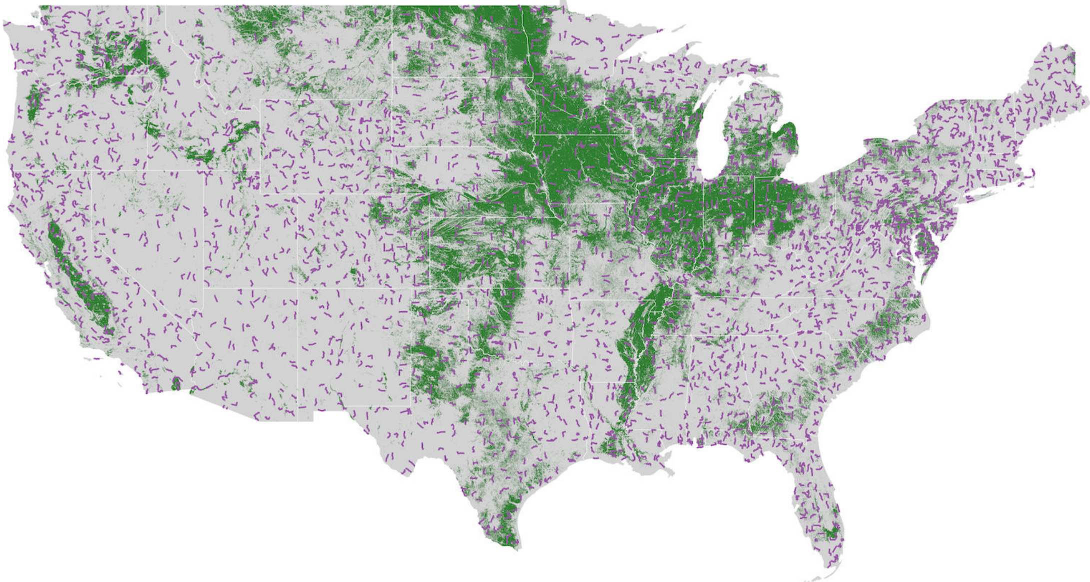


Figure A.2: Pre-anthropocene regional species richness

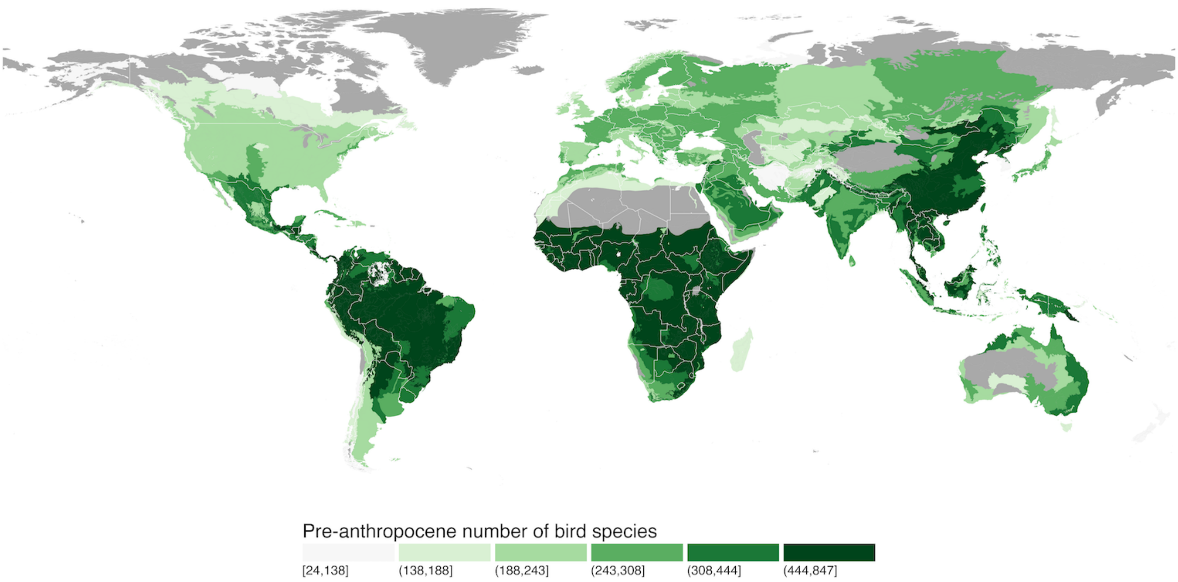
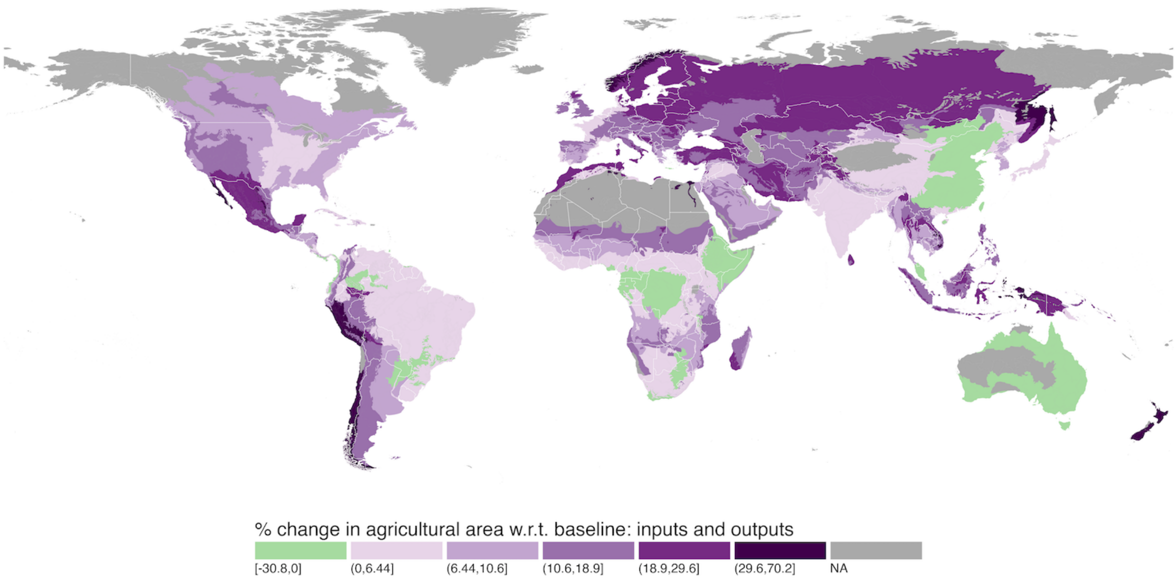
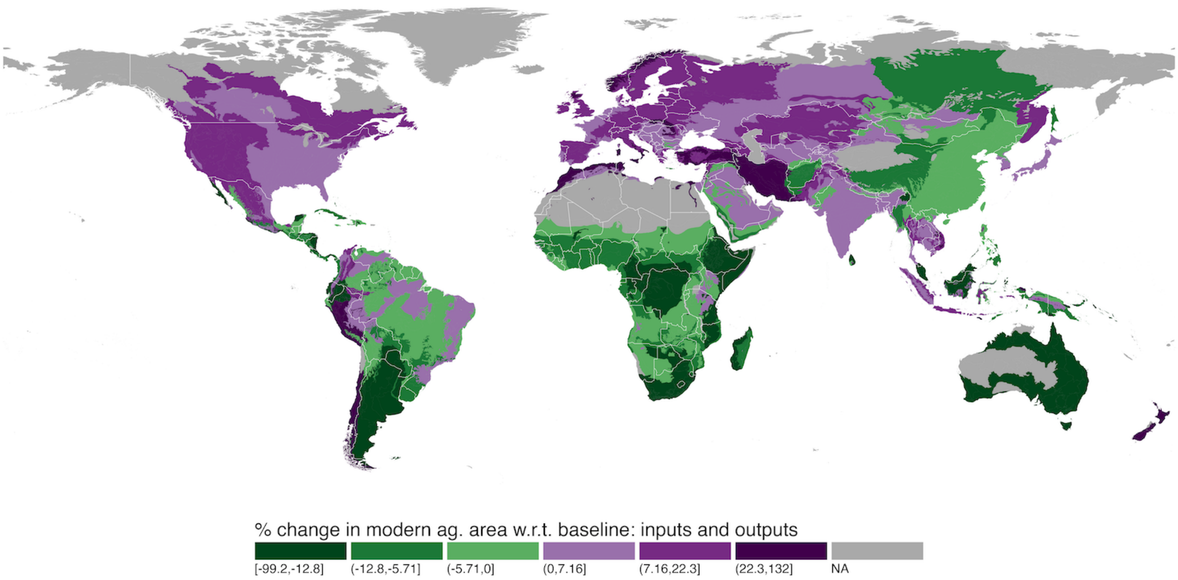


Figure A.3: Impact of setting input and output trade costs to their 1980 level on land-use

(a) Agricultural area



(b) Modern area within agriculture



Notes: Preliminary results without livestock sector and grass production.